On the Shelf Life of Fine-Tuned LLM-Judges

How trained judges age when the generators they evaluate keep getting stronger?

Janvijay Singh[1,2] · Austin Xu[1] · Yilun Zhou[1] · Yefan Zhou[1,3] · Dilek Hakkani-Tür[2] · Shafiq Joty[1]

PROJECT PAGE

01 Motivation

LLM-judges are trained once. Generators keep improving.

- Trained judges score responses from generators they **never saw during training**.
- Does a judge **generalize** under this distribution shift, or does it go stale?

TL;DR

Judges age asymmetrically. Training on weak responses fails on strong ones (up to −6.2%). Training on strong is mostly safe on weak. **Continual training** is the only recipe without a trade-off.

02 Setup

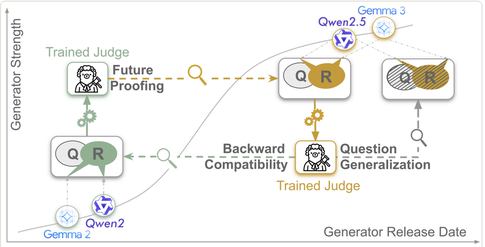


Fig 1. A pairwise judge scores (Q, R_1, R_2). Train-test mismatch decomposes into **R-shift** (weak or strong responses) and **Q-shift** (unseen questions).

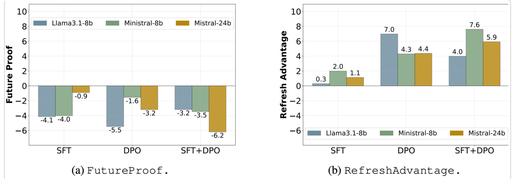
FP - = ages	RA + = recovers	QG - = brittle
FutureProof weak-judge on strong R	**RefreshAdv** gain from retrain	**Q-Generalize** swap Q, hold R
BC + = safe	**CS** + = improves	+ = judge works
BackCompat strong judge on weak R	**CompatShift** strong vs weak baseline	- = judge fails

SCOPE

Backbones: Llama-3.1-8B · Ministral-8B · Mistral-Small-24B · **Recipes:** SFT · DPO · SFT+DPO · DPO-Cont · **Datasets:** DeepScaleR · MMLU-Pro

03 Future-Proof fails

Weak-trained judges do not generalize to strong responses.



(a) FutureProof. (b) RefreshAdvantage.

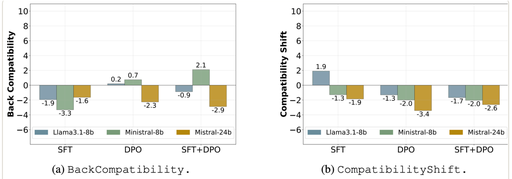
Fig 3. **FutureProof** negative across every backbone × recipe. **RefreshAdvantage** always positive; retraining on strong recovers.

FINDING 1

Every trained judge degrades on newer, stronger responses. DPO extracts the most value from upated stronger responses.

04 Back-Compat holds

Strong-trained judges are roughly safe on older data.



(a) BackCompatibility. (b) CompatibilityShift.

Fig 4. **BackCompatibility** (left) clusters near zero; some judges *improve*. **CompatShift** (right) ≈ −1 to −3 %, much **milder** than FP.

FINDING 2 · ASYMMETRIC GENERALIZATION

Rolling out a new judge is mostly free; a fresh strong-trained judge is a safe drop-in on older generators.

05 Continual beats both

Continual training delivers the best of both worlds, with almost no trade-off.

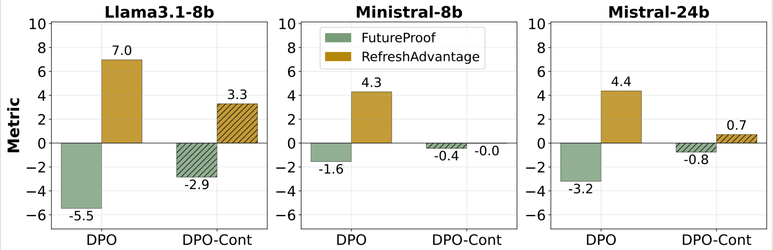


Fig 5. DPO → DPO-Cont on **FP** & **RefreshAdv** . FP magnitude shrinks across all three backbones while refresh gains also reduce.

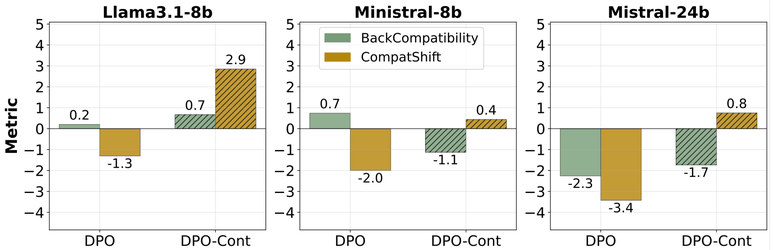


Fig 6. DPO → DPO-Cont on **BC** & **CompatShift**. BC often *improves*; CompatShift flips positive.

FINDING 3

DPO-Cont is the only recipe without a trade-off. Future-proofing improves *and* backward-compat is preserved.

06 Question generalization

Judges drop on unseen questions even with R held fixed.

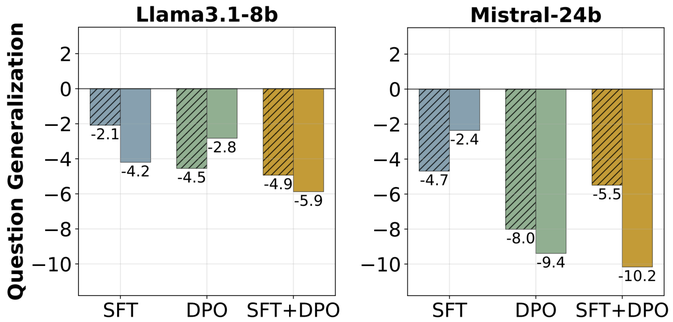


Fig 7. **Mistral-24B**: Q-drift hits hardest on **strong** responses (solid bars). Same asymmetry on Llama3.1-8B & Ministral-8B.

FINDING 4

Response drift isn't the only enemy. Question drift is an orthogonal axis of staleness; diversify **what's asked**, not just **who answers** .

07 Practitioner recipe

IF YOU MAINTAIN A FINE-TUNED JUDGE

- **Retrain with DPO** on outputs from the **latest** generators.
- Fresh, strong-trained judges are **safe to deploy** on older data.
- For evolving stacks, prefer **continual fine-tuning (DPO-Cont)**.
- Diversify your **question pool**, not just the response pool.

TAKEAWAY

Judge staleness is **predictable, asymmetric, and fixable;** if you refresh on the right axis.

Correspondence: jvsingh2@illinois.edu

ICLR 2026 · [1]Salesforce AI Research · [2]UIUC · [3]Dartmouth College

Code/Data: github.com/iamjanvijay/judge-training-analysis